Ergonomics: Technical Worksheet

Grade 11 · Worksheet · About 25 minutes
Work with pen and paper. Use color pencils if you like. You do not need a computer or the internet.

Name: _____

Grade/Section: _____

Date: _____

> **How to use this worksheet**
> - Do the parts **in order** from top to bottom.
> - Each part has a **time** so you know how long to spend.
> - Read every instruction before you write. You can do this on your own.

Part 1 – What you will learn (about 3 minutes)

What you will learn: Ergonomics fits the workstation to the human—posture, reach, lighting, and input device placement—to reduce strain and support longer healthy work.

Do this:

1. **Note one discomfort** you feel after long computer use (neck, wrists, eyes...).

   _____

2. **Circle** ergonomics focus: Fit tools to people · Force people to twist to bad tools

Part 2 – Neutral posture checklist (about 5 minutes)

Targets: feet supported, hips/knees ~comfortable, screen near eye height, elbows ~open angle, wrists neutral—not bent sharply up/down.

> **Real-life scenario:** A laptop on a low desk makes a student crane their neck for hours. A simple riser and external keyboard change the angles.

Discuss with a partner or reflect in writing: Which change helps more first: screen height or chair height? Why might both matter?

Do this: Circle **Higher** or **Lower** risk.

Setup	Riskier?
Screen far below eye level for hours	Higher · Lower
Wrists bent upward while typing	Higher · Lower
Feet dangling without support	Higher · Lower

Part 3 – Input devices and microbreaks (about 5 minutes)

Mouse position near the body's midline, light key force, and **microbreaks** (20–30s every ~20–30 min) reduce cumulative strain. Stretching beats ignoring early pain.

> **Real-life scenario:** After gaming with a low mouse sensitivity requiring big arm sweeps, a student feels shoulder burn.

Critical thinking: How could settings or device placement reduce that motion?

Do this:

1. **Design a microbreak routine** (3 actions, ~30 seconds).

---

2. **Circle** wiser response to wrist ache: Adjust setup + break · Ignore and push harder

Part 4 – Vision and lighting (about 5 minutes)

Reduce glare, match brightness to the room, increase font size when squinting, and use the 20-20-20 idea (every 20 minutes look 20 feet away for 20 seconds) as a practical heuristic.

> **Real-life scenario:** A window behind a monitor creates glare; the student raises brightness to max and gets headaches.

Discuss with a partner or reflect in writing: What environmental fix beats maxing brightness?

Do this: Circle **Yes** or **No**.

Action	Helps visual ergonomics?
Reposition to cut glare	Yes · No
Tiny fonts to fit more code	Yes · No
Periodic far-focus breaks	Yes · No

Part 5 – Station audit sketch (about 5 minutes)

Do this: Sketch your usual station. Mark 3 adjustments (arrows) you can make this week—even low-cost ones.

Prioritize the #1 change and estimate cost/effort.

Part 6 – Check what you learned (about 2 minutes)

Checklist. Put a ✓ in the box when it is true for you.

□ I can define **ergonomics** for computing.

□ I can audit posture and screen height.

□ I plan **microbreaks** and safer input placement.

□ I address glare/vision factors.

□ I can prioritize practical station changes.

When all boxes are checked, you have reviewed the main ideas from this worksheet.

Worksheet on Ergonomics (Technical). Central topic: Ergonomics. Main technology area: Healthy human–computer setup.